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When a Format (Almost) Becomes an Instrument: Eurorack in a Logistical World

ABSTRACT

Eurorack is a format for modular synthesis, a long-running type of electronic sound production that has seen a new wave of interest in the early twenty-first century. This article examines Eurorack as a chimera of putatively analog and digital elements, shaped by a large-scale global logistical system, miniaturized components, and the formatted dimensions of media practice. Embedded in its musical, practical, and logistical contexts, Eurorack helps us model an approach to understanding how technical formats and protocols, transnational flows of materials and labor, and aesthetic practices interweave across the varied scales of modern media.

In Memory of Trevor Pinch

On any number of video sharing platforms today, the right search terms (*ambient, synth, modular, Eurorack*) can lead one down a rabbit hole full of strange sounds and blinking lights. Often attended by houseplants, sometimes by cats, the lights and sounds emanate from electronic instruments called modular synthesizers. The instruments sprout plastic-sheathed wires (patch cables) that their players use to route electrical signals through various sections (modules). The patch is the set of modules and connections between modules that determines the sound—whether that’s a slowly

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pulsating harmonious drone, an evolving sequence of bright staccato bleeps, or a driving drum rhythm. Eurorack is a standard that gives modular synthesists some degree of assurance that the different modules they buy will fit cleanly into a shared case and will properly work with the others. One could purchase two (or more, depending on size) modules, connect them to a power supply hidden from view inside, and screw them to the rails at the top and bottom of the gap. They would then be ready for connection to the other modules in the case (see Figure 1).

Since the mid-2010s, Eurorack has been a flourishing corner of the music instrument industry. In 2012, the synthesizer database on ModularGrid.net listed 876 types of modules available in the Eurorack “universe.”¹ As of May 2024, the site listed 15,472 Eurorack module types, arranged in 483,559 different configurations.² This growth has been something of a surprise to observers both within and without the modular synth world.

The Eurorack synthesizer is a new media technology, and by analyzing it we aim to turn two popular notions in media studies on their heads. First, there is the notion that in the twenty-first century, new media systems draw their distinguishing features mainly from visual interfaces in software or from the ubiquity of screens. Eurorack’s interface elements do not follow the conventions of digital media outlined by scholars such as N. Katherine Hayles, Alex Galloway, Wendy Chun, and Lev Manovich—the screen, the skeuomorph, the mouse, the typewriter keyboard, and even the presets and menus that often structure other recent synthesizers.³ Eurorack modules often embed cutting-edge digital signal processing (DSP) algorithms and other software paired with inexpensive and widely available computing components—or rather, *once* widely available components, as we explain later. At the same time, from the standpoint of mainstream media chronology, Eurorack synthesizers appear as chimeras: drawing on an earlier generation of modular synths from the 1960s and 1970s made by inventors such as Robert Moog and Don Buchla, they combine aspects of twentieth-century instrument panels, server hardware, telephone exchanges, and modern personal media objects such as smartphones and touchscreens.⁴

The chimeric nature of Eurorack technologies points to the need to nuance scholarly discussions of analog and digital technologies. For our purposes, *analog* and *digital* are both analytical terms and actors’ categories. As analytical terms, they refer to specific technological processes. Analog

1. “Modular Grid: Online Modular Synth Database and Rack Planner,” December 7, 2012, accessed January 24, 2022, <https://web.archive.org/web/20121207064952/http://www.modulargrid.net/>.
2. “Plan Your Modular Synthesizer Rack on ModularGrid,” accessed May 29, 2024, <https://www.modulargrid.net>.
3. N. Katherine Hayles, *How We Became Posthuman: Virtual Bodies in Cybernetics, Literature, and Informatics* (Chicago: University of Chicago Press, 1999); Lev Manovich, *The Language of New Media* (Cambridge, MA: MIT Press, 2001); Wendy Hui Kyong Chun, *Programmed Visions: Software and Memory* (Cambridge, MA: MIT Press, 2011); and Alexander R. Galloway, *The Interface Effect* (Cambridge: Polity, 2012).
4. Joseph A. Paradiso, “The Modular Explosion—Deja Vu or Something New?” (paper presented at the Voltage Connect Conference, Berklee College of Music, Boston, MA, March 10–11, 2017).

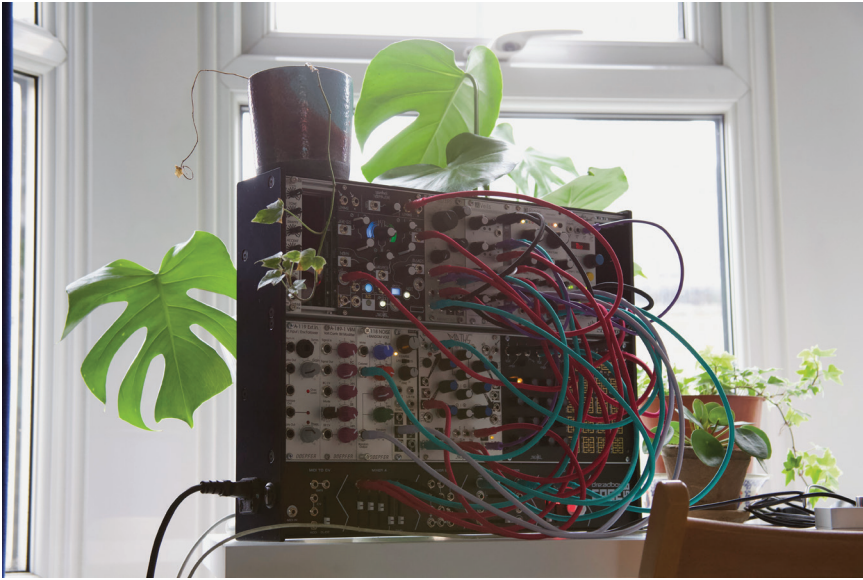


Figure 1. A Eurorack synthesizer, framed by houseplants, combines modules from various manufacturers. In the case surrounding the modules, on the top row, two gaps appear, with rails at their top and bottom. (Photo by John Whiles, 2018).

technologies operate by analogy, as when a certain fluctuation in voltage in an analog synthesizer is analogous to (and controls) a fluctuation in the sound it produces; digital technologies operate through binary code, as when a series of zeros and ones (themselves represented by voltage differences) represent the fluctuations in a sound and can serve as commands for sound production. As actors' categories, digital technologies generally are associated with computation but also with computers and their interfaces; analog is everything that is outside of that (even technologies that are not themselves actually analog technologies, such as a pencil).⁵ As we discuss below, analog synthesizers fell out of popular use in the 1980s and 1990s. But by the early twenty-first century, musicians, journalists, and scholars were already writing about an analog revival characterized by interest in older sound technologies for music-making and playback, an interest mirrored in recent repurposing of older techniques in fields such as photography and printing/bookmaking/zine publishing. Lately, even digital technologies such as flip phones, portable CD players, and iPods have been swept up in so-called analog revivals. Already in 2002, Trevor Pinch and Frank Trocco wrote of the analog revival as musicians' return to "knobs and wires" from digital hardware, computers, and software-based interfaces, but one that was occasioned by "a degree of confusion as to what analog means."⁶ More recently, Laís Akemi

5. See Jonathan Sterne, "Analog," in *Digital Keywords: A Vocabulary of Information Society and Culture*, ed. Benjamin Peters (Princeton, NJ: Princeton University Press, 2016), 31–44.

6. Trevor Pinch and Frank Trocco, *Analog Days: The Invention and Impact of the Moog Synthesizer* (Cambridge, MA: Harvard University Press, 2002), 317, 352.

Margadona has made a similar point about the “new” analog photography, which she calls a “hybrid” photography: “It was forged by film emulsions and silver grains, but it now circulates in the dynamic world of pixels.”⁷ It appears we have been in an analog (or *analogue*?) revival for at least two decades now, which is about to exceed the duration of time that analog technologies were considered obsolete in fields such as music and photography.⁸ Eurorack musicians often mention their desire to move away from music software and desktop and laptop computers as a major reason for exploring and investing in synthesizer hardware. Some will even describe interface elements such as knobs as analog even if the knob is sitting atop an encoder which sends binary values to a microprocessor behind the panel. In other words, digital technology may seem or feel analog-like because it shares interface elements with older analog technology.

Regardless of their actual modes of operation and control, Eurorack’s specific modes of modularity depend on widespread digitization and vast digital networks. Eurorack modules, cases, power supplies, and components are designed in computer programs, often contain microcontrollers, and move through a vast logistical network that is very much a twenty-first-century development, and Eurorack enthusiasts learn their craft through the internet. Even as modular synthesists may crave a more analog-like experience in their interaction with the devices, Eurorack synthesizers both depend on and descend from digital technologies. Eurorack can only be understood in the context of recent logistical changes in the global electronics industry that facilitated its growth.⁹ In other words, the *analog* of the analog revival depends mightily on digital technologies; the *analog* in analog era is not exactly the *analog* in analog revival. As Eurorack matures into a popular mode of production and consumption in electronic music, its logistical entanglements shape the experiences of its users, and perhaps even their musicking.¹⁰

7. Laís Akemi Margadona, “Understanding the New Analogue Photography: Meanings, Methodologies and Aesthetics,” *New Explorations* 3, no. 1 (2023): 4, <https://doi.org/10.7202/1097584ar>.
8. Trevor J. Pinch and Karin Bijsterveld, “‘Should One Applaud?’: Breaches and Boundaries in the Reception of New Technology in Music,” *Technology and Culture* 44, no. 3 (July 2003): 556, <https://doi.org/10.1353/tech.2003.0126>; Andreas Kitzmann and Claes Thorén, “The Modular Journey: Uncovering Analogue Aesthetics in Digital Landscapes,” *Organised Sound* 27, no. 1 (2022): 44–54, <https://doi.org/10.1017/S135577182200022X>; and Claes Thorén et al., “The Hipster’s Dilemma: What Is Analogue or Digital in the Post-Digital Society?,” *Convergence: The International Journal of Research into New Media Technologies* 25, no. 2 (2019): 324–339, <https://doi.org/10.1177/1354856517713139>.
9. Matthew Fuller, *Media Ecologies: Materialist Energies in Art and Technoculture* (Cambridge, MA: MIT Press, 2005); and Deborah Cowen, *The Deadly Life of Logistics: Mapping Violence in Global Trade* (Minneapolis: University of Minnesota Press, 2014).
10. We use Christopher Small’s term *musicking* to treat music as something people make and do, rather than music as a work or a product. In media studies terms, Small’s concept of musicking fuses production studies’ interest with below-the-line labor, infrastructure studies’ interest in large and hidden systems that animate the everyday operation of media, and fan studies’ interest in amateurism and everyday practice. See Christopher Small, *Musicking: The Meanings of Performing and Listening* (Hanover, NH: Wesleyan University Press, 1998), 7–10. For the parallels to Small’s ideas in media studies, see, e.g., Vicki Mayer, *Below the Line: Producers and Production Studies in the New Television Economy* (Durham, NC:

John Tresch and Emily Dolan write that “the fact that the arts and sciences now share so many aspects of their technical infrastructure results in isomorphic logics, shared working strategies, and common imaginaries between them.”¹¹ Eurorack mixes up those logics, working strategies, and imaginaries. As a type of musical instrument or media technology, Eurorack is inseparable in its status as a *format* and a *technical standard*. This makes Eurorack an instructive case for media studies because it represents a future for digital media in which embedded and ubiquitous computing are at least as important as devices that call attention to their computational or digital nature. As Haidee Wasson argues, the analysis of formats can lead to “greater precision and acuity across time and geopolitical boundaries.”¹²

The *euro* in Eurorack is not accidental and should resound in readers’ ears alongside the *euro* in terms such as *Eurocentrism*. If the Eurorack boom is enabled by changes in global logistics and supply chains, that implies both a national order and a geographic politics. In his essay on immobility and obduracy in South African electronic music, Gavin Steingo warns against writing about technological change in a linear fashion—new technological developments introduce new occasions for failure and blockage. But even more so, technological practice changes with context of use. Basic concepts such as mobility take on different meanings, but so, too, do patching and modularity. Steingo rightly argues for more work from and about the Global South, but his point is also important for work like ours that considers media in the Global North.¹³ Most module designers design modules in the Global North, but many of the components for those modules are manufactured in the Global South, and a transnational working class supports the massive shipping enterprises needed to get things wherever they need to be. Eurorack’s very conception emerges from European standards-making operations that depend on Europe as a unifying formation—a set of economic and technological standards meant to lubricate transnational trade for the benefit of countries in the Global North.

While other authors have considered Eurorack from the standpoint of musicians and makers, we aim to describe it as a media phenomenon, while taking on board their accounts of user experience. Thus, this article proceeds in four parts. We consider the digital bases of the so-called analog revival before turning to an analysis of Eurorack as a format and then as a logistical development. We argue that, in many ways, Eurorack’s dependence on logistics and its connection to embedded and ubiquitous computing make

Duke University Press, 2011); Lisa Parks and Nicole Starosielski, eds., *Signal Traffic: Critical Studies of Media Infrastructures* (Urbana: University of Illinois Press, 2015); and Michelle Cho, “3 Ways That BTS and Its Fans Are Redefining Liveness,” *FlowTV*, May 29, 2018, accessed March 17, 2023, <http://www.flowjournal.org/2018/05/bts-and-its-fans/>.

11. John Tresch and Emily Dolan, “Toward a New Organology: Instruments of Music and Science,” *Osiris* 28, no. 1 (2013): 280, <https://doi.org/10.1086/671381>.
12. Haidee Wasson, *Everyday Movies: Portable Film Projectors and the Transformation of American Culture* (Oakland: University of California Press, 2020), 30.
13. Gavin Steingo, “Sound and Circulation: Immobility and Obduracy in South African Electronic Music,” *Ethnomusicology Forum* 24, no. 1 (2015): 103, 113, 120, <https://doi.org/10.1080/17411912.2015.1020823>.

it an important example of new directions in media. Finally, we consider the work of Bamanya Brian, a musician and technologist in Kampala, Uganda (note that Bamanya's name is given with the surname first). "The Afrorack," his synthesizer (as well as his performer name), is built through a combination of global electronics markets and the salvage practices common in Uganda and other countries where Western electronics are often sent to be recycled. Bamanya's work and experience show that Eurorack is a phenomenon that depends on existing international inequalities. But at the same time, his work suggests the promise of a greater consciousness of components' provenance and the value of repair and reuse in robust technological cultures.

To match our chimeric object of study, we have employed hybrid methods in writing this article: participant observation and oral history, walk-through methods, media historiography, and infrastructure studies. Each approach brings something essential. Sometimes stories of technology can best be understood through people's narration of their own experience as users or makers. For this project, Jonathan has been using a Eurorack synthesizer on a regular basis for nine years, designing systems, making music, and regularly screwing up. He has been in regular conversation with other users in workshops, through correspondence, and online; we have also included ethnographic interviews and testimonies of users and makers as primary evidence. Sometimes technologies can best be understood through the provenance of a component or practice; to that end, we analyze the origins of different components and aspects of the Eurorack format as well as the text of published Eurorack standards. Sometimes technologies are best understood through conjunctural analysis; to that end, we aim to situate Eurorack in a twenty-first-century transnational mediascape shaped by unequal labor relations; extraction of natural resources; a system of global logistical flows; and the online remediation of sonic, technological, and musical practices.¹⁴ The story of any format is a hybrid media story. In blending methods, we aim to do a measure of justice to representing this hybridity in the linear form of the journal article.

Gender is also an important methodological and analytic concern. In her study of the emergence of American radio broadcasting, Susan Douglas writes that the post-World War I radio boom was in part the result of a "middle class culture of men and boys" who tinkered with electronic components; the same could be said of some of the stories we tell here.¹⁵ The systemic sexism of electronic music culture, and Eurorack specifically, is absolutely not unique—it is endemic to almost all of the engineering cultures behind

14. Stuart Hall, "Marx's Notes on Method: A 'Reading' of the '1857 Introduction,'" *Cultural Studies* 17, no. 2 (2003): 113–149, <https://doi.org/10.1080/0950238032000114868>; Brian Larkin, *Signal and Noise: Media, Infrastructure, and Urban Culture in Nigeria* (Durham, NC: Duke University Press, 2008); Erkki Huhtamo and Jussi Parikka, eds., *Media Archaeology: Approaches, Applications, and Implications* (Berkeley: University of California Press, 2011); and Jeremy Wade Morris and Sarah Murray, eds., *Appified: Culture in the Age of Apps* (Ann Arbor: University of Michigan Press, 2018).

15. Susan J. Douglas, *Inventing American Broadcasting, 1899–1922* (Baltimore: Johns Hopkins University Press, 1987), 22.

media practices and media technologies, from the radio cultures Douglas studied to computer science and artificial intelligence.¹⁶ There have been long-standing debates about the reasons, effects, and dimensions of sexism within Eurorack culture. Until 2021, the name for the main website where people could gather to learn about and discuss Eurorack—alongside other modular synthesizer formats—was named Muff Wiggler. Mike McGrath, the site's founder, originally intended the name as an ironic joke: a mash-up of the names of two Electro-Harmonix guitar pedals, a popular fuzz distortion pedal called the Big Muff Pi, and the Wiggler, a vibrato pedal.¹⁷ While McGrath actively worked to make the place welcoming to all users, he was not entirely successful, and Muff Wiggler existed alongside other online forums with sexist names such as Gearslut, which became the major source for information on studio and recording equipment online—and changed its name in 2021.¹⁸ This is not just a matter of language or naming, though; rather, the name changes are symptomatic of broader concerns about participation and opportunity. For the past few years, organizations such as Keychange have pushed, with some success, for electronic music festivals to build truly gender-diverse programming.¹⁹ Large companies such as Ableton include explicit language about gender discrimination in their programming for technologists and artists alike. We have tried to be sensitive to this dynamic. Most of the older actors in our story are cisgender men; their stories are valuable evidence as they provide access to some of the technological stories we wanted to analyze. But it's important to remember these men

16. On the relation between sexism in music tech and sexism in tech, see Georgina Born and Kyle Devine, "Music Technology, Gender, and Class: Digitization, Educational and Social Change in Britain," *Twentieth-Century Music* 12, no. 2 (2015): 135–172, <https://doi.org/10.1017/S1478572215000018>. See Jennifer S. Light, "When Computers Were Women," *Technology and Culture* 40, no. 3 (1999): 455–483; Venus Green, *Race on the Line: Gender, Labor and Technology in the Bell System, 1880–1980* (Durham, NC: Duke University Press, 2001); and Gregory J. Downey, *Telegraph Messenger Boys: Labor, Technology, and Geography, 1850–1950* (London: Routledge, 2002).

17. In US white English slang, "muff" can refer to a woman's pubic hair and "pie" to a vagina. The pedal was manufactured by a US company.

18. As with the Muff Wiggler case, the founder of Gearslut claimed that he was being ironic and self-deprecating, but at some point one has to ask why the same so-called joke is repeatedly made. In 2014, Asha Tamirisa, a media artist and theorist, described the many forms of implicit and explicit sexism she encountered on the forum, ranging from images and discussions of women's bodies to overt misogyny. Her article in turn led to a long thread on the forum discussing sexism and is part of ongoing arguments among participants—some taking explicitly feminist positions and others taking explicitly anti-feminist positions, which seem to sometimes still be tolerated by moderators despite the name change. See Asha Tamirisa, "Muff Wiggler: Sexism in Audio Cultures," April 17, 2014, accessed June 7, 2022, <https://bluestockingsmag.tumblr.com/post/82994060875/muff-wiggler-sexism-in-audio-cultures>; and "Is Muff Wiggler Offensive?," accessed January 31, 2022, <https://modwiggler.com/forum/viewtopic.php?t=488>.

19. See Keychange, accessed January 15, 2022, <https://www.keychange.eu>. For more on sexism, feminism, and audio cultures, see Tara Rodgers, *Pink Noises: Women on Electronic Music and Sound* (Durham, NC: Duke University Press, 2010). Jonathan explores this issue further in a forthcoming essay, "The Sexual Politics of Music Equipment." On other technological cultures, see Mar Hicks, *Programmed Inequality: How Britain Discarded Women Technologists and Lost Its Edge in Computing* (Cambridge, MA: MIT Press, 2017); and Christina Dunbar-Hester, *Hacking Diversity: The Politics of Inclusion in Open Technology Cultures* (Princeton, NJ: Princeton University Press, 2020).

were moving in and through homosocial worlds buffeted by their normative status. As we move toward examples from the present, we have tried to present a more diverse—or at least gender-ambiguous—image of Eurorack makers and practitioners. Although this article is mostly about the present and very recent past, much more historical work remains to be done. As writers such as Tara Rodgers have shown, women and gender-diverse people have been written out of synthesizer history, but it does not mean they were not present.²⁰

A DIGITAL ANALOG REVIVAL

It was February 2012. Finding Tom Oberheim's shop involved heading out to a strip mall/office park off an interstate in Northern California and then locating it among a set of nondescript businesses. Once inside, there was a room lined with shelves, a few tables, and synth pioneer Tom Oberheim. He was in the middle of assembling a group of Synthesizer Expander Modules (SEMs), the instrument that made him famous in the 1970s; empty faceplates and parts were set out on a table in an orderly fashion. Jonathan had come that day to interview him as part of a project on the history of signal processing. But as with all research, sometimes the story shifts when you get to the site. On the shelves were plain brown boxes with labels such as Mouser and Digi-Key.²¹ They signaled that a change had taken place in the circuitry that would go behind those faceplates, one that bound this classic synthesizer design up in a far-flung and fast-paced new world of electronics manufacturing.

Oberheim had recently gone back into business after being out of business for two decades. His synth company had been forced into bankruptcy in mid-1985. While standard lore is that the success of the Yamaha DX7, a digital synthesizer with a variety of useful preset sounds for working keyboardists, killed the analog synthesizer business, the story is a little more nuanced. Oberheim said that because of the DX7's success, his business was selling even more instruments. The problem came from two different directions: competition from *all-in-one sampling workstations* with sequencing and composition options and simple market overextension. At that time, Oberheim's company was building a complex synthesizer called the Matrix-12. They had ordered a large cache of parts that, coupled with high labor costs, ran them into a cash flow problem. A local bank foreclosed on US\$1 million of his debt. His lawyer then double-crossed him, bought the rights to his company, and later sold his name to Gibson Guitars. In other words, digitization mattered, but logistics and cutthroat capitalism destroyed his company, not the DX7.

In 2008, Oberheim accepted a speaking invitation at a Red Bull Music Academy in Barcelona (mostly for the free trip). While there, he met a group of young electronic musicians who were enamored of his instruments; they

20. Tara Rodgers, "Tinkering with Cultural Memory: Gender and the Politics of Synthesizer Historiography," *Feminist Media Histories* 1, no. 4 (2015): 5–30. <https://doi.org/10.1525/fmh.2015.1.4.5>.

21. All details from this section are from Tom Oberheim, in discussion with Jonathan Sterne, February 2012.

persuaded him to get back into the business. But the business itself had changed in the intervening decades. When he decided to build the SEM again, he no longer had the schematics. That wasn't a problem, because they were now freely available on the internet. He also switched to surface-mounted components, which made manufacturing considerably easier: "It's expensive to have things assembled the old way where the resistor's leads go through the holes in the board."²² Surface-mount technology saved Oberheim costs in labor and time.

In the 1970s, Oberheim was eager to leave the SEM behind to build bigger, more complex synthesizers, adding polyphony, patch memory, automatic tuning, and a host of features. In 2012, he was back to building one of his first and simplest products, in his words, "regurgitating old Oberheim. And I'm really happy doing it."²³ He wasn't alone. By 2012, companies like the one run by his friend Dave Smith were building new programmable versions of analog synthesizers with a small group of employees. He also noticed the then-emerging Eurorack business: "the new modular companies; the thing that's kind of fun for me is . . . this just continual announcement of new little synth modules because it just helps me" by acting as a kind of free promotion for his work.²⁴ The logistics and standards problems that had driven Oberheim out of business in 1985 had fallen away thanks to an elaborate system including flexible-scale parts suppliers, distributed manufacturing, surface-mount technology, direct-to-consumer sales and marketing, and a set of interlocking communities of enthusiasts on the internet, as well as discounted labor in the Global South. These same global and logistical factors, coupled to the Eurorack format, propelled a new wave whose significance to the synthesizer world was beginning to be felt.

Eurorack is the latest chapter in a long history of sound synthesis, which has involved a constant negotiation around what exactly constitutes a musical instrument and, more generally, the legitimacy of different kinds of technologized musical practice.²⁵ Modularity has been a foundational principle to the synthesizer since its debut as a marketed musical instrument. As Pinch and Trocco explain, Moog (and his contemporary Buchla) embedded two "crucial ideas" in their first designs in the 1960s: "first, that voltage control could be applied to an electrical musical instrument, and, second, that the instrument could consist of discrete modules (oscillators, amplifiers, envelope generators, and, later on, filters) that could be wired together in a variety of ways and controlled by the output voltages of the devices themselves."²⁶ This principle of individuating sonic processes into electrical inputs and outputs—and charting different signal voyages among them—would persist

22. Oberheim, discussion.

23. Oberheim, discussion.

24. Oberheim, discussion.

25. Christopher Latina, "Machine Listening Eurorack Module" (master's thesis, Georgia Institute of Technology, 2016); and Yanto Browning, Stephanie Hutchison, and John McCormick, "Performing High Voltage: Full Body Gestural Control of Eurorack Modular Systems" (paper presented at the 4th International Conference on Movement Computing [MOCO 2017], London, June 28–30, 2017).

26. Pinch and Trocco, *Analog Days*, 28.

as the imaginative and technical range of those processes steadily expanded, even as designers began to sculpt them in software.²⁷

Few if any observers predicted the resurgent interest in modular synthesis that now attends the Eurorack explosion.²⁸ The general arc of synthesizer history in the United States, Western Europe, and other wealthy countries seemed headed in different directions. While, as Oberheim noted, the 1980s and 1990s produced large integrated synthesis and sampling workstations, music production software boomed in the 1990s and 2000s, and by the 2010s, most industry predictions seemed to highlight the promise of touchscreens, mobile technologies, and analog hardware with digital control.²⁹ Even in 2012, it was not clear to Oberheim or either of the authors that Eurorack was about to be a big thing. Chroniclers of sound technology have offered competing explanations for the phenomenon of the Eurorack format and what it has to teach adjacent fields. They highlight Eurorack's flurry of esoteric, enthusiast technical development and the local and distributed social structures that support it, as when Ezra Teboul locates Eurorack module production among various "technocommunal" practices that audio electronics experimentation germinates.³⁰ Farley Miller centers the consumerist appeal that, in spite of the do-it-yourself (DIY) values at play in synth rig customization, animates a growing stream of forum communities and YouTube demonstrations whereby "the heterogeneity of Eurorack module design and manufacture feeds back into the scene's ethos of consumption."³¹ Eliot Bates, who characterizes the social apparatus around Eurorack as a "milieu culture," focuses on the material affordances that drive interest away from software interfaces and toward those made up of hardware modules.³² At the same time, the popularity of software synthesis helped pave the way for Eurorack. Bates identifies six "modes of interfaciality" at play among synth modules and musicians.³³ This diversity of ways to interface in Eurorack presents a contrast with software synthesizers that rely on mice, keyboards, and touchscreens; it

27. Jonathan Sterne and Tara Rodgers, "The Poetics of Signal Processing," *differences* 22, no. 2–3 (2011): 31–53, <https://doi.org/10.1215/10407391-1428834>.

28. Eliot Bates, "The Interface and Instrumentality of Eurorack Modular Synthesis," in *Rethinking Music through Science and Technology Studies*, ed. Antoine Hennion and Christophe Levaux (London: Routledge, 2021), 170–188; and Paradiso, "Modular Explosion."

29. A much more fulsome unspooling of this history can be found in Mike D'Errico, *Push: Software Design and the Cultural Politics of Music Production* (New York: Oxford University Press, 2022). Outside the Global North, the interest in MIDI hardware and computer-driven production was also a going concern; the equipment was just relatively more expensive and harder to come by. See, for example, Louise Meintjes, *Sound of Africa! Making Music Zulu in a South African Studio* (Durham, NC: Duke University Press, 2003); Andrew J. Eisenberg, "Soundtracks in the Silicon Savannah: Digital Production, Aesthetic Entrepreneurship in Nairobi, Kenya," in *Music and Digital Media: A Planetary Anthropology*, ed. Georgina Born (London: University College London Press, 2022), 46–90.

30. Ezra Teboul, "Electronic Music Hardware and Open Design Methodologies for Post-Optimal Objects," in *Making Things and Drawing Boundaries: Experiments in the Digital Humanities*, ed. Jentery Sayers (Minneapolis: University of Minnesota Press, 2017), 177.

31. Farley Miller, "Co-consuming a 'Fun Addiction': Buying, Desiring, and Using Eurorack Together Online," *Sound Effects: An Interdisciplinary Journal of Sound and Sound Experience* 9, no. 1 (2020): 17, <https://doi.org/10.7146/se.v9i1.112699>.

32. Bates, "Interface and Instrumentality," 172.

33. Bates, 178.

spurs “the emergence of a vibrant manufacturer/designer ecosystem and a proliferation of numerous variants of each technological object type.”³⁴

Interest in hardware synthesizers is a countercurrent to the power and ease of use of software synthesizers. Hardware devices often had analog and digital components. They followed many different paths. Various “silicon luthiers” and “DIY music technologists” built highly customized closed instruments; companies such as Dave Smith Instruments and later Dreadbox built hardware synthesizers; and others, such as Doepfer, pursued open, modular systems.³⁵ But all of these stories—Tom Oberheim, silicon luthiers, and especially Eurorack—are cases of small capitalism relying on big capitalism.

FORMATS, STANDARDS, INSTRUMENTS, MEDIA

All the above analyses depend on Eurorack modules working and working together. In this light, the formatted and standard-specified dimension of Eurorack has some explanatory power for how contemporary music synthesis and its modular technologies work. A format is a mixture of aesthetic rules, social practices, and technical practices that define the sound, look, feel, and operation of a media technology. Among other things, a format is a set of rules for how a media form relates to itself as well as other media technologies and forms. Like technological standards, formats are recursive: they tend to refer to other media formats and standards rather than an external reality they aim to reproduce.³⁶ Moreover, formats show that a single medium, or a single technology (like sound synthesis), may not have a single governing operational logic; it may in fact have several, each of which are taken up by different social milieus. Media theories can obscure this multiplicity through an overemphasis on internal technics. For instance, Friedrich Kittler claims that analog and digital systems had fundamentally different operational logics, unique “media-technical flows of commands and addresses.”³⁷ If we were to seek out a fundamental operational logic for Eurorack in Kittler’s style, we would miss out on the important differences between Eurorack and other forms of sound synthesis, which lie not deep in the bowels of its circuits, but in its status as a formatted technology, circulating through webs of logistical and cultural practices. As Georgina Born argues, “to understand the mutual mediation of music and digital media means reading out from

34. Bates, 181.

35. Ezra J. Teboul, “Silicon Luthiers: Contemporary Practices in Electronic Music Hardware” (master’s thesis, Dartmouth College, 2015); and Lauren Elizabeth Flood, “Building and Becoming: DIY Music Technology in New York and Berlin” (PhD diss., Columbia University, 2016).

36. As Jonathan has written, “Format denotes a whole range of decisions that affect the look, feel, experience, and workings of a medium. It also names a set of rules according to which a technology can operate. In an analog device, the format is usually a particular utilization of a mechanism. An old record player may play back a variety of formats such as LP, 45, 78, while a tape deck might only take compact cassettes. In a digital device, a format tells the operating system whether a given file is for a word processor, a web browser, a music playback program, or something else. Even though this may seem trivial, it can open out to a broader politics, as an administrative issue across platforms.” See Jonathan Sterne, *MP3: The Meaning of a Format* (Durham, NC: Duke University Press, 2012), 8.

37. Friedrich Kittler, “Non-Linear Oscillators & Computer Motherboards” (2010), lecture, European Graduate School Video Lectures, January 5, 2011, YouTube video, 59:21, accessed June 7, 2022, <https://youtu.be/CxIHwCnVYIE>.

the digital-musical object or event (MP3 file, track, album, performance, work, genre) to not only the material but the social, cultural, political, legal, ideological and aesthetic processes and relations in which it is entangled.”³⁸ This process is even more important for technologies like Eurorack that are both digital and not fully digital.

In his essay “Reformatting Media Studies,” Axel Volmar argues for a turn to formats that “shift scholarly attention from questions of mediation to those regarding the interoperability of technologies, media-related practices, labor chains, and exploitation chains as well as collective forms of technological innovation.”³⁹ He notes that media scholars have failed to foresee the rise of social media because of misplaced assumptions about the specificity of media technologies. Something similar happened with accounts of Eurorack: Eurorack exists at the intersection of media technology, musical instrument, interface, and signal processor. It is made up of materials and elements that circulate through a global traffic in electronic components and materials, and it is inconceivable without a vast digital infrastructure that suffuses everything from how modules are designed and function to their sizes and shape and interface to how people acquire, dispose of, learn, and transform them. It is both analog and digital; it is at once smoothly interoperable and not totally reliable.

Eurorack partakes of several categories of format as outlined by Volmar in his taxonomy: it defines size and shape; it is diagrammatic in providing parameters for interface elements such as knobs, sliders, touchscreens, joysticks, plugs and cables; and it includes (incomplete) protocols for signal form, content, and control.⁴⁰ But Eurorack is different from other media formats and instruments in that makers’ and users’ relationships to it are in part shaped by an attention to the format *as a format*—a system from which instruments can emerge and a system that can fail to fully concretize as a coherent instrument. The shape and texture of the format allows actually existing (and imagined) Eurorack synthesizers to be continually unfinished, revisable, and reformable. Perhaps the best example of how this works can be found on ModularGrid, which is a website for users to lay out different potential configurations of modules in their system through dragging and dropping images of modules into images of racks; they can then arrange and rearrange them (see Figure 2). Users can also name and save different configurations for easy lookup later.

38. Georgina Born, “Introduction: Music, Digitisation and Mediation—for a Planetary Anthropology,” in Born, *Music and Digital Media*, 25.

39. Axel Volmar, “Reformatting Media Studies: Toward a Theoretical Framework for Format Studies,” in *Format Matters: Standards, Practices, and Politics in Media Cultures*, ed. Marek Jancovic, Axel Volmar, and Alexandra Schneider (Lüneberg, Germany: Meson Press, 2020), 29.

40. Volmar’s taxonomy includes size-and-shape formats, such as the shapes of blank sheets of paper or instrument panels; diagrammatic and structural formats, which prepare surfaces for their use as media, such as typesetting and graphic design; encoding and data formats, which govern signal flow and decoding, such as MP3 audio or NTSC video; metaphorical formats, which is a German locution to describe capability, courage, or resolution; and narrative and processural-event formats, such as Top 40 radio or self-help books.



Figure 2. A Eurorack mock-up on ModularGrid, assembled by Jonathan Sterne.

As Figure 2 illustrates, in addition to displaying the layout of the case, ModularGrid shows cumulative list price, power consumption, and physical depth of modules. The last two categories are particularly important, since one could purchase a module too deep to fit into some racks or exceed some racks' available power. Users can save collections of modules, providing a kind of Pinterest for synthesizer components. These designs can be used to plan out and build actual systems or to sketch out different configurations of the same modules (see Figure 3). In addition to being a planning resource, ModularGrid is a museum of imaginary instruments, a large number of which do not and cannot work on their own. As Miller points out, Eurorack is a “never-ending instrument,” where the experience of design and customization is integral to the experience of using the instrument.⁴¹

Miller is right that commodity fetishism plays a role in modular synthesis, though this is true of most Western instruments. Assuming the means, space, money, motivation, and availability, one can keep buying guitar pedals, bass pickups, trumpet mouthpieces, violin bows, and flute head joints in an effort to modify an instrument's sound or playing technics, until one or more of those resources runs out. But each of those instruments is, when working correctly, a predictable part of a system.⁴² Into the first years of the twenty-first century, modular synthesists tended to purchase single systems from single manufacturers, where there might be some opportunities for customization but not an expectation that the instrument would be constantly remade.

41. Miller, “Co-consuming,” 16, 11. The consumerism of Eurorack is widely noted by users. And Eurorack instruments tend to cost more than fixed-architecture synthesizers, though considerably less than, for instance, many brass instruments or woodwinds.

42. Steve Waksman, *Instruments of Desire: The Electric Guitar and the Shaping of Musical Experience* (Cambridge, MA: Harvard University Press, 1999); see also Meintjes, *Sound of Africa!*

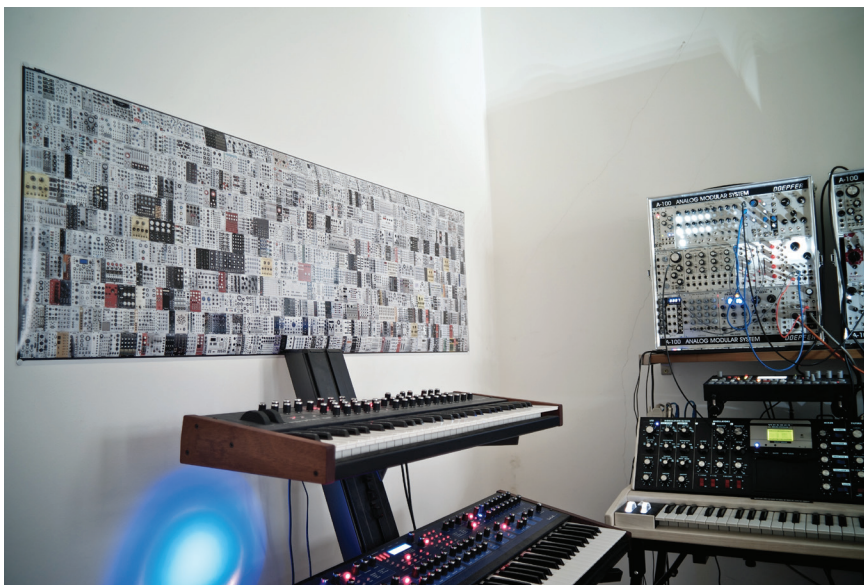


Figure 3. A Eurorack synth in a Doepfer-branded case (top right) and a poster showcasing ModularGrid's selection of Eurorack modules (top left) accompany other synths. (Knut, ModularGrid .Net Poster, January 25, 2015, <https://www.flickr.com/photos/instantknut/16177193058/>.)

While each might have options for customization, the basic function and sound generating parameters of the instrument were set in advance; modification was not a condition of their instrumentality. Of course, there is a long history of desire for new instruments that exceed the boundaries of what was available. Thomas Patteson argues this is a constitutive dimension of twentieth-century avant-garde composition.⁴³ And playback technologies are frequently transformed by use that does not follow designer intent—the first DJs essentially hacked their turntables. Within a few months of the CD player's commercial release, musicians were experimenting with creating music from amplifying its malfunctions.⁴⁴

In some sense, Eurorack is an expansion and domestication of these tendencies, alongside hardware hacking and DIY practices that grew in popularity in the first decade of the twenty-first century and the software aesthetics that grew up alongside the popularization of software programs such as Max/MSP, where users were expected to design entirely novel instruments as part of their practice (even if in practice they often borrow from one another).⁴⁵ But as it works now, Eurorack has a fundamentally different

43. Thomas Patteson, *Instruments for New Music: Sound, Technology, Modernism* (Oakland: University of California Press, 2015).

44. Tricia Rose, *Black Noise: Rap Music and Black Culture in Contemporary America* (Hanover, NH: Wesleyan University Press, 1994), 62–97; Mark Katz, *Groove Music: The Art and Culture of the Hip-Hop DJ* (New York: Oxford University Press, 2012), 43–69; and Caleb Kelly, *Cracked Media: The Sound of Malfunction* (Cambridge, MA: MIT Press, 2009), 214–215.

45. Joe Snape and Georgina Born, "Max, Music Software, and the Mutual Mediation of Aesthetics and Digital Technologies," in Born, *Music and Digital Media*, 220–266.

kind of openness. Not only can Eurorack musicians define for themselves the operational parameters of their instruments, apart from the relative minority who purchase a single manufacturer system, but they *must* go through a process that used to be reserved for a small minority of musicians. That is, they must design their own instrument before playing it. Fueled by the format, standard, and economies behind it, Eurorack's openness allows it never to be finished. The fact that instrument building is so fully integrated with musicking among Eurorack musicians means that any given instrument is formatted to be unfinished (or only temporarily finished) at any given time. This is why the term *journey* is used by musicians and critics alike in narration of their practice.⁴⁶

Eurorack is both a format and a standard. Eurorack is a standard because it is built according to a set of written-down rules—in this case, a pair of written standards, *A-100 Construction Details* and *Technical Details A-100*, which specify size, shape, connectivity, nomenclature, and a few other aspects of the format.⁴⁷ The rules are neither legal nor binding. While to be sold, a Eurorack module must conform to certain electrical safety and interference standards set by states and international consortiums, its conformity to the Eurorack standard itself is not legally required. However, the choice to conform or not has all sorts of social and economic implications. If a manufacturer releases a synthesizer module in another standard, for instance Moog Unit (MU) or Buchla, it can be installed in one of those systems, but not a Eurorack system, meaning that there is potentially a much smaller market for the module. Both standards and formats define interoperability, which is crucial for modular synthesizers: users want to know in advance whether a module from a particular maker will work with their existing systems. This is no different from interoperability in another field, whether we are talking about USB connectors for computers or consumer electronics, national differences in electrical outlets, or digital file formats. Generations of scholars in media studies and science and technology studies have noted that standards are highly political, and interoperability is the axis through which these politics are exercised.⁴⁸ When companies set standards, it is usually to gain advantage in an industry by defining the terms of interoperability. When governments set them, it is usually to settle industrial debates and promote an economic vision. When international bodies set them, it is to help facilitate or manage international commerce and collaboration. But makers—whether they are individuals

46. Miller, "Co-consuming"; and Kitzmann and Thorén, "Modular Journey."

47. Doefer, *A-100 Construction Details*, https://doepfer.de/a100_man/a100m_e.htm; and Doefer, *Technical Details A-100*, accessed January 31, 2022, https://doepfer.de/a100_man/a100t_e.htm. Interestingly, Dieter Döfer got his start in the synthesizer and component import-export business, so the genesis of Eurorack is directly tied to the history of logistics.

48. François Bar, Michael Borrus, and Richard Steinberg, "Islands in the Bit-Stream: Charting the NII Interoperability Debate" (working paper, Berkeley Roundtable on the International Economy, University of California, Berkeley, 1995), 21–22, https://brie.berkeley.edu/sites/default/files/wp_79.pdf; Geoffrey C. Bowker and Susan Leigh Star, *Sorting Things Out: Classification and Its Consequences* (Cambridge, MA: MIT Press, 1999); Sterne, *MP3*, 130; and Dylan Mulvin, *Proxies: The Cultural Work of Standing In* (Cambridge, MA: MIT Press, 2021).

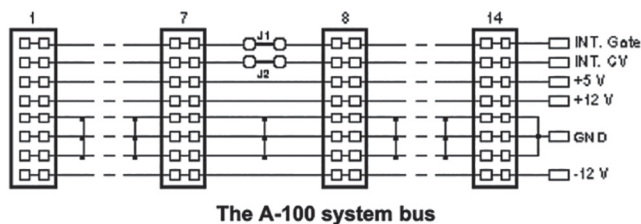


Figure 4. A diagram of the A-100 System Bus. (Doepfer, *Technical Details A-100*, https://doepfer.de/a100_man/a100t_e.htm.)

or companies—can ignore or abandon standards just as easily when it is to their advantage to do so.

These kinds of standards are common for formats, which often emerge from “media of cooperation” that help manage “the relations between different users or user groups, their goals, and their practices.”⁴⁹ Eurorack was outlined by Doepfer, a private company—really one person to begin with, Dieter Döpfer—but published openly. As with the earlier versions of modular synthesizers discussed above, it was originally intended more as a set of instructions for people to build their own Doepfer systems than as a standard for other manufacturers to follow. But other manufacturers used it. The fact that Doepfer did not charge others to use their standard is one of the major reasons for the format’s success, though it is not unique to it, since many modular synthesizer systems are built on open standards. Like other formats, Eurorack originated from temporary and non-public working documents that were conceived to support the work of building something, in this case a synthesizer.

Parts of the standard are usually followed carefully, such as the horizontal pitch (HP) measurements laid out in *A-100 Construction Details*. Almost all modules on the market come with a HP measurement (1hp = 5.08mm = 1/5"). All modules expect a +12/-12-volt power supply and connect with a 16-pin connector on a powered bus board. There are also specifications for spade connectors that connect the power supply with the bus board, which distributes power to modules.

And yet lurking in this documentation is evidence that most Eurorack modules do not fully follow the standard. Figures 4 to 6 show Doepfer’s system bus, which distributes both power and control within a system of modules. But most manufacturers do not use the internal CV or gate bus, so the top four pins in Figure 4 are mostly unused. Additionally, there is no commonly agreed-upon standard for the 5-volt power source supply (the third row of pins in Figure 4). Some modules use it; some do not. Some systems have it; some do not. There are no instructions about electrical shielding for digital modules—a recurring problem in multi-manufacturer systems and one that we have heard firsthand in the shape of unexpected buzzes, whines, and hums. In fact, it is possible that even though any commercially produced Eurorack modules must meet electromagnetic interference requirements

49. Volmar, “Reformatting Media Studies,” 38.



Figure 5. An A-100–style system bus, showing plugs and a single power cable connected. (Photo by Carrie Rentschler.)



Figure 6. A single Eurorack module, showing its conformity to the 3U rack standard, with its plug extending down to the A-100–style system bus. (Photo by Carrie Rentschler.)

to which all commercial electronics are subject, assembled systems may not meet those standards.⁵⁰

Formats often refer to other formats, and standards often refer to other standards; they contain “nested references.”⁵¹ *A-100 Construction Details* refers to the nineteen-inch rack format from *IEEE Standard for Mechanical Core Specification for Microcomputers* (IEEE Standard 1101-1987), which uses the term *Euroboard* frequently and has considerably more detail about electrical tolerances than the Eurorack standard. Today’s range of Eurorack cases still refers to it, but manufacturers do not exactly follow it. Doepfer’s modules were all 3U tall (5.25" or 133.35mm) and fit into nineteen-inch rows (84hp), as illustrated in Figure 6. Today’s systems can vary in size in all sorts of ways. They can be considerably wider or narrower and may have any number of 3U rows stacked on top of one another, and there are at least two different 1U (1.75") Eurorack standards in circulation that are not fully interoperable with one another (Intellijel and Pulp Logic). The standard also says nothing about inter-case connections, another common source of noise and problems for users.

Power in Eurorack is a major issue, and a bit more dangerous than just plugging a USB cable into a phone or a plug into a wall. The A-100 standard advises to “check very carefully that it is connected so that the colored marking on the ribbon cable is at the bottom of the bus connector, and it is pushed fully home, not at a slight angle. Failure to check this may result in the module’s instant destruction as soon as the power is turned back on!”⁵² The red stripe on the cable in Figure 5 is the “colored marking” to which Doepfer refers. This seems like a simple admonition and good advice. But the power standard is an example of how Eurorack is an incomplete format. These absences—around electrical tolerances and inter-case connections—make the standard seem even more open than it actually is in practice. Eurorack lore is full of stories of people plugging modules in the wrong way, of incompatible or underpowered supplies, of modules that are unshielded and bleed (unwanted) noise into other parts of the system to which they are connected. As Volmar argues, formats “show people how to use technologies and how not to use them,” but that does not mean they are necessarily complete.⁵³

Being a modular format, Eurorack is in another sense deliberately incomplete; it is designed for musicians to repeatedly disconnect and reconnect patch cables to make new signal routes and to install and remove modules to alter the set of patchable elements. Gilbert Simondon uses the term *concretization* to describe convergence and co-shaping among a machine’s internal parts.⁵⁴ Concretization also characterizes musical instrument design: minute changes in internal parts play out over time in accordance with properties such as resonance or string tension in acoustic instruments or impedance and signal flow in electronic instruments. The resulting, stabilized

50. “Roundtable: Low Volume Manufacturing in Eurorack, hosted by Jeff Lee and Heidi Chan” (roundtable at Patch-Up! conference, Toronto Metropolitan University, April 2, 2022).

51. Mulvin, *Proxies*, 18.

52. Doepfer, *A-100 Construction Details*, 5.

53. Volmar, “Reformatting Media Studies,” 39.

54. Gilbert Simondon, *On the Mode of Existence of Technical Objects*, trans. Cecile Malaspina and John Rogove (Minneapolis: University of Minnesota Press, 2016).

instrument takes on a coherent set of qualities in the view of its players. In contrast, the Eurorack synthesizer remains “abstract” in Simondon’s terms, where “each theoretical and material unit is treated as an absolute . . . ; integration into an ensemble in this case raises a series of so-called technical problems that must be resolved.”⁵⁵ Eurorack reserves for the performer the work of connecting and integrating its various modules into an ensemble that will function as an instrument. The Eurorack standard helps to reduce the range of “so-called technical problems” (in Simondon’s words) that might obstruct this ensemble-making, but it hardly eliminates them.

Because of its openness, it is hard to say that Eurorack has a sound; it sounds like synthesizer music. Acknowledging that any canon would be hotly debated by artists and listeners, we suggest that some classic Eurorack albums could include Keith Fullerton Whitman’s *Generator* (Whitman, 2010), which was made with a Doepfer and Plan B system; Alessandro Cortini’s *ACMN 1&2* (Make Noise Records, 2012), which was made with a Make Noise Shared System; and Caterina Barbieri’s *Patterns of Consciousness* (Important Records, 2017), which was made with a hybrid system consisting of an Orthogonal Devices Sequencer, a Verbos Harmonic Oscillator, and software effects from Ableton Live. These albums prominently feature analog sound generation and Western tonality; they prominently lack a standard keyboard polyphony that might characterize music made by modern fixed-architecture polyphonic keyboard synthesizers and samplers. Emily A. Sprague’s *Water Memory* (Sprague, 2017) uses a truly hybrid Eurorack system from many makers, alongside a hardware synth and software reverbs, and sounds more like other sample-based compositions. While these are all examples of work by solo artists, Eurorack sometimes features in ensemble contexts. For instance, Lichens (Robert Aiki Aubrey Lowe) performs with voice and modular synthesizer and has collaborated with a range of artists. Metal band REZN’s *Solace* (Big Cartel, 2023) features a Eurorack player. Eurorack synths are routinely used in sound design for film, television, and games, and they appear in sound art installations (and sometimes as decoration in music videos and advertisements). Electronic musicians originally famous for work with other kinds of technology—such as Scanner, Amon Tobin, and Lady Starlight—have taken to composing and performing with Eurorack in recent years. As with any instrument, musicians often purchase equipment that artists who influence them have used. Influence does not amount to sonic determination, though, as Eurorack manufacturer Metasonix satirically warns by hand-writing *You will not sound like NIN* on the back of a circuit board for an amplifier module—a reference to the band Nine Inch Nails’ use of their products.

Eurorack does have somewhat stronger associations with genres that have appeared at the same time and from overlapping cultural interests. Rik Adriaans explains that “the modular synthesizer revival has today become commonly associated with a specific genre which embraces the midcentury sensibility of a modular music without keyboards: generative ambient music.”⁵⁶ This genre, sometimes dubbed “houseplant ambient” in light of a visual trope

55. Simondon, 27.

56. Rik Adriaans, “Synthesizer Sceneries: Performing Natural Music on Social Media” (paper presented at the European Association of Social Anthropologists, Belfast, July 27, 2022).

in the exquisitely tidy videos that showcase it, “is essentially a soothing and meditative music that incorporates elements of randomness and feedback into the signal path of a composition, which as a result becomes ever-evolving and will never sound exactly the same.”⁵⁷ It hews close to what George Lewis called “Eurological indeterminacy” when discussing the work of American composer John Cage.⁵⁸ Cage developed his concept of indeterminacy as a compositional approach that left substantial elements of a piece’s performance up to chance or the performers’ interpretation. A large set of Eurorack modules provide controlled randomness for sequencing and triggering sound generators. Indeterminacy was a radical idea in twentieth-century art music, where greater control for composers was a widely held value. But as Lewis points out, Cage and other composers worked to distinguish their practice from improvisational practices in American jazz.⁵⁹ There was thus a racial politic behind the indeterminacy concept, and inasmuch as Eurorack musicians draw on the white avant-garde, rather than jazz improvisation traditions, that history remains alive today. But genres are cultural rather than technical phenomena, and other practitioners have made Eurorack the vehicle for sounds that have a more agonistic relationship with Eurological musical norms. For instance, the percussiveness and timbral variety that characterize Bamanya Brian’s compositions for his Afrorack project sound nothing like ambient soundscapes: same formats, same protocols; different people, different music.

COMPONENTS, LOGISTICS, AND OTHER SEAMS

For all its openness to different interfaces, sounds, and internal technologies, the Eurorack phenomenon has been subject to trends in the larger global electronics industry. Since the time we began researching this project, rippling disruptions from the COVID-19 pandemic created shortages in electrical components and especially in the microcontroller chips that many DSP-based modules used. Several prominent module makers cited this “chip shortage” as a main or major reason for deciding to cease Eurorack production.⁶⁰ Manufacturing and distribution are at least as important to understanding Eurorack as interface elements and interoperability. Attention to logistics emphasizes the vital importance of software-based systems to the analog revival—from the micro level of components and modules, to the mezzo level of design software, to the macro level of global shipping and supply chains.

Single-employee operations are not uncommon among the hundreds of companies offering Eurorack modules, cases, and accessories. But even such small-scale production takes place as a confluence of global streams

57. Adriaans.

58. George E. Lewis, “Improvised Music after 1950: Afrological and Eurological Perspectives,” *Black Music Research Journal* 22 (2002): 215–246.

59. Lewis.

60. Paul Schreiber, “Synthesis Technology FUTURE Plans,” *Mod Wiggler*, May 29, 2022, accessed January 20, 2023, <https://www.modwiggler.com/forum/viewtopic.php?t=262562>; Crystal Koe, “Eurorack Synth Maker Hexinverter Closing within the Next Year Due to Global Supply Chain Issues,” *MusicTech* (blog), September 14, 2022, accessed January 20, 2023, <https://musictech.com/news/industry/eurorack-synth-maker-hexinverter-closing-global-supply-chain-issues-chip-shortage/>; and Daniel Mulder, “Re: Livestock Electronics Has Shut Down,” *Mod Wiggler*, October 28, 2022, accessed August 20, 2024, <https://modwiggler.com/forum/viewtopic.php?p=3850629#p3850629>.

for electronics manufacture, including services from component supply to circuit board printing that in many cases make it possible for an individual to coordinate the production of hundreds or thousands of modules. These resources were not always available to businesses that cater to niche creative sectors, and they did not become available inevitably. Developments in electronics and logistical industries, as well as specific projects that formed pathways between industrial and creative contexts, helped determine the timing and the social makeup of the Eurorack phenomenon.

In her role as the operator of Mutable Instruments, the French technologist Émilie Gillet experienced firsthand several scenes and stages of Eurorack's production side. When Gillet started Mutable Instruments, she produced DIY kits for stand-alone synthesizers rather than the preassembled Eurorack modules that would later make up most of the highly regarded company's offerings. She describes the transition from DIY kits as a movement from component sourcing flows "connected to what I would call the maker scene—stores like Sparkfun or Adafruit, PCB [printed circuit board] services like PCBCart, consumer-facing parts sellers like Mouser or Digikey, laser cutting services"—into an operation that "increasingly looked like a traditional hardware company."⁶¹ An older and more local network of manufacturing contractors, some with ties to France's telecom, aircraft, and medical equipment industries, supported Mutable Instruments in its Eurorack phase compared to its initial years; yet it was precisely the emergence (starting in the 1990s) of that newer, more dispersed network targeting small-batch and creative electronics makers that would lay essential groundwork for Eurorack.

When Mutable Instruments switched from DIY kits to preassembled modules, it also switched between types of electronic circuitry. The kits had featured through-hole components, an older style whose electrical leads can be threaded by hand through holes in a circuit board (or a breadboard for prototyping); the Eurorack modules sported the smaller, flatter-lying components used in surface-mount technology (SMT). SMT, which over the course of the 1980s and 1990s overtook through-hole as the industrial standard approach for attaching electrical components to circuit boards (see Figures 7 and 8), followed and propelled two longer trends in electronics: miniaturization and automation.⁶² Miniaturization has long been tied to ideas both of progress and of stigma; the drive to hide hearing aids and their batteries and Gordon Moore's misnamed "law" that chip density would double every eighteen months both play a part in its history.⁶³ High-density integrated circuit (IC) chips became crucial components for sound synthesizers as more complex and portable models proliferated in the decades before Eurorack.

Perhaps even more than its reduction in component sizes, SMT's significance for the electronics industry stems from its capacity for automated assembly, which gave rise to new manufacturing intermediaries that could

61. Émilie Gillet, email message to Jonathan Sterne, February 11, 2022.

62. Ray P. Prasad, *Surface Mount Technology: Principles and Practice*, 2nd ed. (New York: Chapman and Hall, 1997).

63. Richard Maxwell and Toby Miller, *Greening the Media* (New York: Oxford University Press, 2012), 28; and Mara Mills, "Hearing Aids and the History of Electronics Miniaturization," *IEEE Annals of the History of Computing* 33, no. 2 (February 2011): 24–45, <https://doi.org/10.1109/MAHC.2011.43>.

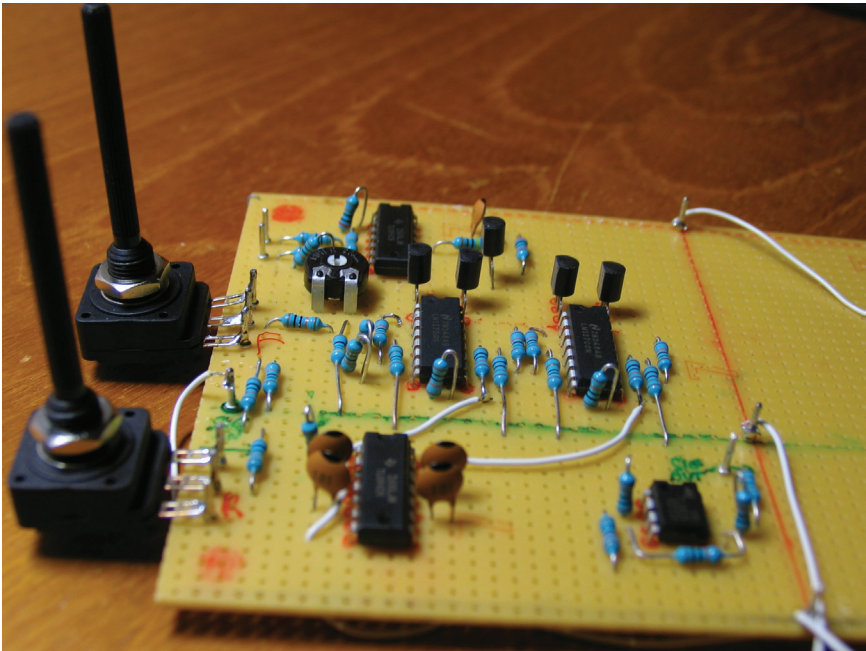


Figure 7. ICs and other through-hole components are assembled on a breadboard to form a DIY module. (John Cloudy, Flickr.com, 2005, shared under CC BY-NC-ND 2.0.)

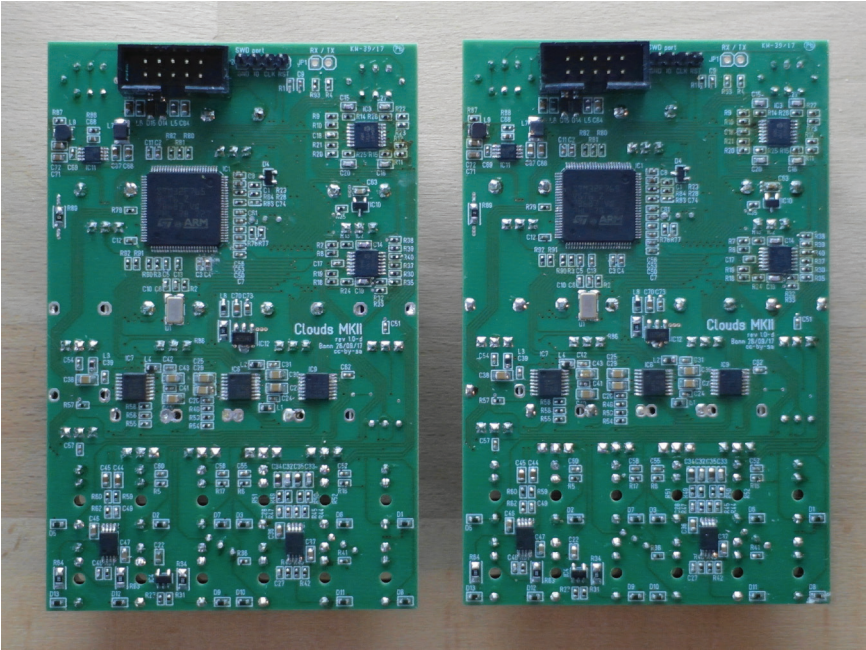


Figure 8. A pair of PCBs for a prototype Mutable Instruments module shows surface-mounted components. (Photo by Émilie Gillet, 2021.)

produce fully assembled circuits for both large- and hobbyist-scale clients.⁶⁴ Automation, like miniaturization, has been crucial to manufacturing industries from the 1940s onward. Both trends would appear in Eurorack: miniaturization made it possible to fit complex signal processing circuits onto slender circuit boards that can slot into shallow cases, and automation allowed for efficient and scalable contract manufacturing.

Alongside claims for its efficiency, automation implies labor conflict. Work formerly done by people is done by machines. But as with most automation, it often involves new kinds of discounted labor to maintain and assist the machinery.⁶⁵ For instance, Oberheim explained that by reviving his old circuit designs in a mostly SMT production design, he was able to reduce labor cost while retaining spots for particular higher-end components that he saw as essential.⁶⁶ In other words, fewer people had to be paid—or at least, fewer people had to be paid high wages. SMT's flexible design matched a flexible scale. A small shop could theoretically meet a sudden swell in demand without incurring financial risk or compromising on the custom touches through which builders distinguish their work.

This scalar flexibility in circuit assembly would have had limited impact without corresponding flexibility in supply chains for the components themselves. In the United States, electronic parts distributors Digi-Key and Mouser both rapidly expanded their operations from the 1990s into the next decade, with Mouser opening a Shanghai office in 2008.⁶⁷ These firms make low-cost components, often produced in countries with low wages, available to customers around the world, where interest in analog synthesis was solidifying through dedicated forums and trade conventions. Digi-Key and Mouser distinguished themselves by fulfilling tiny orders of a few components at a time in addition to the large wholesale orders that corporate clients would place.⁶⁸ As was the case with SMT, which relies on circuit design and simulation software in place of through-hole's greater capacities for hands-on prototyping, software played a key role in turning a more complex technical arrangement into a manageable cost saver for small-scale creators. Anne Pasek and Nicole Starosielski, writing about design and supply chain management software in the boutique garment industry, note that “software allows designers a familiar and personalized sense of creative control; it tightens the links in the industrial work process through virtual production management. This is not simply automation, but a reconfiguration of skill, coordination, and communicable knowledge within a wider system of creative production

64. Jerry Pickert and Charles L. Hutchins, “Surface Mount Technology—the Future of Electronic Assembly,” *IEEE Circuits and Devices Magazine* 3, no. 2 (March 1987): 33–41, <https://doi.org/10.1109/MCD.1987.6323236>.

65. Sumanth Gopinath, *The Ringtone Dialectic: Economy and Cultural Form* (Cambridge, MA: MIT Press, 2013), 59–60, 78–79; and Mary L. Gray and Siddharth Suri, *Ghost Work: How to Stop Silicon Valley from Building a New Global Underclass* (New York: Houghton Mifflin Harcourt, 2019), 38–64.

66. Oberheim, discussion.

67. Mouser Electronics, “Mouser Electronics Continues Asia Expansion with New Shanghai Customer Service Center,” press release, September 30, 2008, accessed December 16, 2022, <https://www.mouser.com/pdfdocs/PressRoomshanghaienglish.pdf>.

68. Jean Kumagai, “A Day in the Life of Digi-Key,” *IEEE Spectrum* 51, no. 2 (February 2014): 50–56.

and virtual control.”⁶⁹ The same can be said of electronic design automation (EDA) software that helps Eurorack module designers prototype in software, cheaply build test models, and then outsource manufacturing with a high degree of precision.

As in the garment industry, new tools played important roles in making supply chains and their accompanying technology available to Eurorack builders. But unlike the garment industry (so far), some of these key tools involved the widespread embedding of computation directly into the boutique products themselves. By Gillet’s reckoning, a key precipitator for the Eurorack boom was the Arduino project, which in 2005 presented a toolkit that artists and designers could use to integrate small computational components into physical objects.⁷⁰ Arduino, which built on other projects, created an articulation point between the electronics industry and an ascendant creative technologist or “maker” movement.⁷¹ Gillet credits Arduino with “drastically lowering the barrier to entry for embedded programming.”⁷² Like the terms *physical computing* and *ubiquitous computing*, *embedded programming* denotes a design approach where microcontrollers perform dedicated functions within special interactive housings. Eurorack modules like the ones Mutable Instruments produced are perfect examples: each module hosts a carefully tailored bit of digital signal processing along with its appropriate knobs, switches, inputs, outputs, and indicators. But they work well alongside analog-only modules. Rather than enforcing a single underlying technical logic, physical computing propelled the Eurorack format by expanding its range of functions.

Eurorack’s debt to physical computing and the logistical networks that subtend it suggests a larger point about logistics and creative systems. Physical or ubiquitous computing has often been the site of aspirations for an aesthetic of seamlessness: for large-scale infrastructural terrains of quietly operational and interlinked computational objects and for a supposedly seamless user experience. This has led to calls to acknowledge and even valorize “seamfulness”—the opposite of seamlessness—that this aesthetic hopes to cover over.⁷³ Eurorack renders this tension visible, touchable, and audible. The format shows its seams on its surface in the boundaries between modules, in the different design aesthetics of different manufacturers, and in the very presence of patch cables. On the surface, it might be comparable to a garment practice like visible mending, in that it aestheticizes its seams.⁷⁴

69. Pasek and Starosielski, “Fiber, Fingers, and Code,” 475.

70. Jentery Sayers et al., “Between Bits and Atoms: Physical Computing and Desktop Fabrication in the Humanities,” in *A New Companion to Digital Humanities*, ed. Susan Schreibman, Ray Siemens, and John Unsworth (New York: Wiley, 2015), 1–21.

71. Hernando Barragán, “The Untold History of Arduino,” 2016, accessed December 16, 2022, <https://arduinohistory.github.io>. Arduino built on a project called Wiring that in turn adapted the Processing framework for graphical media art programming.

72. Gillet, email message.

73. Sarah Inman and David Ribes, “‘Beautiful Seams’: Strategic Revelations and Concealments,” in *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (New York: Association for Computing Machinery, 2019), <https://doi.org/10.1145/3290605.3300508>.

74. Anna König, “A Stitch in Time: Changing Cultural Constructions of Craft and Mending,” *Culture Unbound: Journal of Current Cultural Research* 5, no. 4 (2013): 569–585, <https://doi.org/10.3384/cu.2000.1525.135569>.

But unlike visible mending, Eurorack practice tends to obscure labor, contingency, and wear. The format requires imagining a kind of seamlessness by allowing different modules, designed and assembled across many settings and with any number of conflicting ideas about musical signal processing, to fit together in a neat grid and to pass sound back and forth with one another. The seamlessness that allows a Eurorack synth to function as an instrument, in other words, occludes seams of the sort Pasek and Starosielski spotlight: “the points where materially, geographically, and logistically different processes come together to form an industry or supply chain.”⁷⁵ They also point out that the disappearance of those seams requires the imaginary disappearance of the “low-wage, racially Other workers” who labor at them.⁷⁶

Over time, Eurorack use will inevitably crack the format’s impression of seamlessness. Despite the format’s best efforts, a module will draw too much power or send higher voltages than its neighbor can handle, throwing smooth signal flow into chaos or silence. A module will get plugged in incorrectly and let off a puff of toxic smoke. As a new media object, the Eurorack synthesizer reveals a distinct set of opportunities for makers, musicians, and audiences to probe those more immediate technical seams and their connection to logistical seams, even if that opportunity is rarely taken up. The act of connecting two modules with a patch cable represents the fleeting, temporary coherence of messy, global, and heterogeneous parts and processes.

Depending on one’s position in global supply chains, Eurorack can look very different from the picture we have painted so far. The career of Bamanya Brian illustrates this well. A guitarist and synthesist in Kampala, Uganda, he acquired some internet fame after posting a January 14, 2019, YouTube video where he performed on a self-built modular synthesizer.⁷⁷ With manufactured Eurorack modules out of his financial reach, he built his own modules out of recycled electronic components, calling it Afrorack as an answer to Eurorack. Afrorack references Eurorack from its social and technical margins; musical creation becomes a way of pushing the seams that constitute it into the foreground.

A few months ago I dipped my feet into the world of modular synths. The entire African continent has [only] a handful of shops selling Euro rack modules which is really sad. Also, Euro rack modules are overly priced for someone in my context. My only option was to go the DIY route. I looked up schematics and started modifying some circuits and building others from scratch. All parts in my modular were sourced locally. In a few months I have learnt so much about electronics and analog synths and I have built what is an almost

75. Pasek and Starosielski, “Fiber, Fingers, and Code,” 478.

76. Pasek and Starosielski, 477.

77. Bamanya Brian, “Africa’s First Home Made Modular Synth,” Afrorack, January 14, 2019, YouTube video, 9:48, accessed June 8, 2022, <https://youtu.be/ov2sEfUv1XY>. This was not his first video. Bamanya had been posting on YouTube for about a year, starting with a DIY guitar amp and DIY solar charge controller using a 555 IC, a particularly popular IC chip because it is relatively simple and can work with a very wide range of voltages, making it hard to break and suitable for many different kinds of projects.

decent modular synth. I believe this is Africa's first home made modular synth. I hope this video inspires someone out there to build their own synth. Feel free to ask any questions and please share and subscribe.⁷⁸

Within months, Bamanya was drawing attention from manufacturers—French hardware and software maker Arturia posted an interview with him about his use of one of their products—and electronic music journalist Peter Kirn wrote that “[t]he modular phenomenon, nice as it is, has often been confined to deep pockets and particular geographies. So I’m sure this story will speak to anyone who feels isolated and/or on too tight a budget for some of the gear out there.”⁷⁹ But Kirn’s universalist language leaves aside the specificities of Bamanya’s practice. He is not laboring in solitude, but as part of a scene: as an associate of the Nyege Nyege Tapes label and part of Kampala’s vital music culture. Bamanya gets his parts three ways: “I recycle as much as I can” (for instance, from discarded power supplies), which is “a good practice to recycle electronics because of the habit of throwing away perfectly working tech”; he shops for components in the used component markets in downtown Kampala; and he uses eBay—“come on, I don’t live in a jungle . . . there is no special status for me because I live in Africa”—as well as Mouser and Amazon. Some things are more expensive in Kampala, like Arduinos, which he buys online to save money.⁸⁰

Consider the similarities and differences in Oberheim’s or Gillet’s stories. The relative position of the Ugandan schilling in global currency markets, as well as Africa’s position in global economies, puts the standard Eurorack system well out of financial reach for most Ugandan musicians. DIY is the only way into modular synthesis for someone like Bamanya. Secondhand electronics take on a different meaning in the Ugandan context, compared with countries in the Global North. Like other African countries, Uganda has e-waste scrap yards and imports a large quantity of used electronics. This leads to a large market in used devices and components, as well as a culture of reuse and recirculation and, finally, a longer use-life for electronic devices, something that scholars of African media have regularly noted.⁸¹ While DIY Eurorack enthusiasts may choose to use recycled parts because they want to, recycled parts are part of the Afrorack synthesizer because they must be. At the same time, Afrorack (the person) partakes of the same global logistical economies as makers in the Global North. Afrorack offers a glimpse of a version of modern modular synthesis that goes beyond simply referencing other technological regimes and practices: he actively incorporates and reuses them. This is not to romanticize the North’s dumping of its electronic waste on the Global South, but to argue that Bamanya’s synthesis highlights the user-level seams of the patch and the transnational seams sewn by logics and

78. Bamanya.

79. Peter Kirn, “Afrorack and Africa’s First DIY Modular Synth Make Some Wild, Delicious Grooves,” *CDM* (blog), January 30, 2020, accessed June 8, 2022, <https://cdm.link/2020/01/afrorack-first-african-modular/>.

80. Bamanya.

81. Both Larkin and Steingo discuss this aspect of media practice in Africa.

legacies of colonial capitalism. Afrorack provincializes Eurorack by showing how Eurorack operates on top of enduring inequalities between the Global North and the Global South.

CONCLUSION

Our analysis of Eurorack draws attention to emergent processes in the world of media that may be more instructive for future research than fealty to legacy media forms and old ideas of medium specificity. The near future of digital media is one that spans scalar registers, where components, formats, standards, infrastructures, protocols, models, and logistical systems matter as much as anything like a medium in older parlance. This means that we may have to give up the search for fundamental media logics, considering instead mixtures and contradictions among different clusters of logics, practices, and registers. For instance, we have shown that embedded and physical computing can be taken up in a wide variety of directions. That means we cannot declare Eurorack or modular synthesis to have an inherent political meaning. Whether we are talking about its infrastructural politics, labor politics, or gender politics, Eurorack has both radical and regressive elements. A technology cannot have a single conjuncture; it has many. We cannot know ahead of time where and how ideology functions in a given media form. Maybe the interface is all about demystification while the back end is all about mystification, and maybe the revelation of a seemingly more equitable manufacturing and distribution structure conceals other politics. In practice, maybe a format or standard is more open, or more closed, than it seems in theory. Maybe the apparent freedom of an individual designer or artist depends on a tight and inflexible logistical structure and poisonous web of resource extraction. Or perhaps the converse is the case.

In this way, Eurorack looks like one future for media technologies: computing embedded in devices that do not feel like computers to their end users and the media operations inside technologies that do not feel like media to their end users. Instruments once seemed more distinct from the world of media than they currently are; the same is true for many other technologies. In his book *The Human Uses of Human Beings*, Norbert Wiener uses elevators and gun sights as examples of cybernetic systems that depended on feedback.⁸² Today those same systems might well serve as examples for how the end user experience is conditioned by a mesh of analog, digital, and physical processes, themselves conditioned by formats, standards, protocols, and logistics that reach across scales. What could a term such as *medium specificity* even mean in such a setting? The context is both much bigger and much smaller; formats, standards, practices, and techniques may matter at very different scales than the term *medium* suggests. To describe and analyze this world, we need to simultaneously attend to the component dimensions of legacy media systems such as film, television, or sound recording and to draw our analytics and metaphors from beyond them, to the wide, varied world of material

82. Norbert Wiener, *The Human Uses of Human Beings: Cybernetics and Society* (Cambridge, MA: Riverside Press, 1950), 12.

cultures and technological and cultural practices—from artworks to grey literature and machinic and institutional forms that now exist around us.

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